and a strategy called Suspicious Tit for Tat. Suspicious Tit for Tat is technically nasty, but it is not *very* nasty. It behaves just like Tit for Tat itself after the first move, but—this is what makes it technically nasty—it does defect on the very first move of the game. In a climate entirely dominated by Tit for Tat, Suspicious Tit for Tat does not prosper, because its initial defection triggers an unbroken run of mutual recrimination. When it meets a Tit for Two Tats player, on the other hand, Tit for Two Tats's greater forgivingness nips this recrimination in the bud. Both players end the game with at least the 'benchmark', all C, score and with Suspicious Tit for Tat scoring a bonus for its initial defection. Boyd and Lorberbaum showed that a population of Tit for Tat could be invaded, evolutionarily speaking, by a *mixture* of Tit for Two Tats and Suspicious Tit for Tat, the two prospering in each other's company. This combination is almost certainly not the only combination that could invade in this kind of way. There are probably lots of mixtures of slightly nasty strategies with nice and very forgiving strategies that are together capable of invading. Some might see this as a mirror for familiar aspects of human life.

Axelrod recognized that Tit for Tat is not strictly an ESS, and he therefore coined the phrase 'collectively stable strategy' to describe it. As in the case of true ESSs, it is possible for more than one strategy to be collectively stable at the same time. And again, it is a matter of luck which one comes to dominate a population. Always Defect is also stable, as well as Tit for Tat. In a population that has already come to be dominated by Always Defect, no other strategy does better. We can treat the system as bistable, with Always Defect being one of the stable points, Tit for Tat (or some mixture of mostly nice, retaliatory strategies) the other stable point. Whichever stable point comes to dominate the population first will tend to stay dominant.

But what does 'dominate' mean, in quantitative terms? How many Tit for Tats must there be in order for Tit for Tat to do better than Always Defect? That depends upon the detailed pay-offs that the banker has agreed to shell out in this particular game. All we can say in general is that there is a critical frequency, a knife-edge. On one side of the knife-edge the critical frequency of Tit for Tat is exceeded, and selection will favour more and more Tit for Tats. On the other side of the knife-edge the critical frequency of Always Defect is exceeded, and selection will favour more and more Always Defects. We met the equivalent of this